Proposition 8: remaining fixes

12 October 2026 | only outstanding actions | original checklist numbers retained

NOW = work for 13 October under temporary continuity (C); RED = proof of (C), left for later; BOTH = split work. **(C):** $d\Psi : (0,\infty) \to \mathcal{L}(\mathbb{R}, L^2(\mathbb{R}))$ is continuous in operator norm (Lemma 7, line 774).

Only (C) is deferred. The uniform remainder in Proposition 8 is to be proved tomorrow.

Notation: $a = h_{min}$, $b = h_{max}$, $\delta = \delta_{N,j}$, $L = b^{1-r}$, $J = [a,b]$.

7. Finish the remainder inequalities. [NOW]

Order the endpoints at line 755 for negative τ; change the enlarged-supremum equality at line 756 to $\leq$. Use $|\tau_{N,j}|$ in any nonzero quotient at line 833; handle $j=i_N$ separately, with increments and remainders zero. Define M_0 to cover both derivative bounds (item 10).

Source: thesis lines 755-756; 833; 841; 855.

8. Complete the red-remainder proof. [BOTH]

NOW: At lines 778-779 use operator norms and $\leq$ when enlarging the supremum to all close pairs. Define μ_J on fixed J. Under (C), uniform continuity gives $\mu_J(s) \to 0$ [1], and the integral remainder gives $\|r_h(\tau)\|_2 \leq |\tau|\mu_J(|\tau|)$. Make the closing sentence at line 781 conditional on (C). At line 844 retain $\mathbf{C_g M_0 |\delta|^2}$; justify the uniform grid bounds and choose $a_N(A)$ independently of j:

$\mu_J(s) = \sup\{\|d\Psi(x)-d\Psi(y)\|_{op} : x,y \in J, |x-y| \leq s\}$.
$C_g = \frac{1}{2}(1-r)\max\{a^{1-2r}, b^{1-2r}\}$.
$a_N(A) = L\mu_J(LA/\sqrt{\lambda_N}) + C_g M_0 A/\sqrt{\lambda_N} \to 0$.

RED: Only the analytic proof of continuity at line 774 remains for later. The identity $\|d\Psi(s)-d\Psi(h)\|_{op} = \|\Psi'(s)-\Psi'(h)\|_2$ reduces it to L^2 continuity; the Gaussian dominated-convergence route is available [3].

Source: thesis lines 774-781; 835-845; 852-871; Zeidler [1]; DLMF MVT/Taylor [2]; Hunter [3].

10. Establish derivative bounds and energy properties under (C). [NOW]

Continuity (C) and compactness give a finite bound below for both $\|\Psi'(h)\|_2$ and $h^{1-r}\|\Psi'(h)\|_2$. They also give continuity of Q_r through its definition. Prove $Q_r > 0$ from its Gaussian Fourier formula and $P_n(0)=1$; record the Fourier derivative if needed. Then connect a_N to the existing b_N and exponential-error bounds.

$M_0 = \max\{1,L\} \sup_{h\in J} \|\Psi'(h)\|_2 < \infty$.

Source: thesis lines 358-366; 841; 855; 871; 874. Detailed energy formula in the review.

11. Finish the Riemann-sum details. [NOW]

State $\varepsilon_N \to 0$ and define $m_N = \lfloor A/\varepsilon_N \rfloor$. The widths at line 891 use right endpoints: use $[-(m_N+1)\varepsilon_N, m_N\varepsilon_N]$, whose endpoints tend to -A,A. Account for the endpoint pieces. Fix the δ index in $t_{N,1}$ at line 887; endpoint divergence supplies all local indices.

$\varepsilon_N = ((b^r-a^r)/r)\sqrt{\lambda_N}/(N-1) \to 0$.

Source: thesis lines 797; 887-894; DLMF Eq. (1.4.18) [2]. The tail and final squeeze need no new argument.

13. Make notation consistent. [NOW]

Use one δ and τ notation, μ_J, and $S^{(N)}$. Fix $r_{N,j}$, η_{Nj}, ρ_{Nj}, m_n, and unindexed ε. Use $r_h(\tau)$ at line 738; remove duplicated wording at line 722. Use operator norms for operators and L^2 norms for functions. State uniformity over local j for each fixed A.

Source: thesis lines 722; 738; 778-779; 828-894.

14. Qualify the interpretation and Perron claim. [NOW / SEPARATE RESULT]

State the row-sum comparison asymptotically for fixed interior limiting bandwidths. Use Q_r, or explain $\beta=2(1-r)$. Slow change refers to $v=h^r/r$; Q_r is half the squared speed. The Perron-concentration claim needs a separate theorem outside Proposition 8. Resolve the Laplace TODO as a writing decision.

Source: thesis lines 714; 909-918.

Target for tomorrow: finish Proposition 8 and its interpretation conditional on (C). Only the continuity proof in Lemma 7 is deferred.

All corrections are original derivations from the thesis. Sources for the general tools:

[1] Zeidler (1995), Applied Functional Analysis, §4.4, Prop. 4(ii), p. 241; Cor. 5, p. 242; Example 3, p. 240.

[2] NIST DLMF §1.4, Eqs. (1.4.11), (1.4.18), and §1.4(vi).

[3] Hunter, Measure Theory, Thm. 4.24, p. 49.

Thesis: /Users/iljasukhanov/MASTERARBEIT/Abschlussarbeit_2021/LaTeX/chapters/5_bandwidth_selection.tex